

Bias-Variance Tradeoff

- The expected error on an unseen sample x can be decomposed as:

$$\begin{aligned} E[(y - \hat{f})^2] &= E[(f + \varepsilon - \hat{f})^2] \\ &= E[(f + \varepsilon - \hat{f} + E[\hat{f}] - E[\hat{f}])^2] \\ &= E[(f - E[\hat{f}])^2] + E[\varepsilon^2] + E[(E[\hat{f}] - \hat{f})^2] \\ &\quad + 2E[(f - E[\hat{f}])\varepsilon] + 2E[\varepsilon(E[\hat{f}] - \hat{f})] + 2E[(E[\hat{f}] - \hat{f})(f - E[\hat{f}])] \\ &= (f - E[\hat{f}])^2 + E(\varepsilon^2) + E[(E[\hat{f}] - \hat{f})^2] \\ &\quad + 2(f - E[\hat{f}])E(\varepsilon) + 2E(\varepsilon)E(E[\hat{f}] - \hat{f}) + 2E[E[\hat{f}] - \hat{f}](f - E[\hat{f}]) \end{aligned}$$

Bias-Variance Tradeoff

$$\begin{aligned} &= (f - E[\hat{f}])^2 + E[\varepsilon^2] + E[(E[\hat{f}] - \hat{f})^2] \\ &= (f - E[\hat{f}])^2 + \text{Var}[y] + \text{Var}[\hat{f}] \\ &= \text{Bias}[\hat{f}]^2 + \text{Var}[y] + \text{Var}[\hat{f}] \\ &= \text{Bias}[\hat{f}]^2 + \sigma^2 + \text{Var}[\hat{f}] \end{aligned}$$

- Hence the derivation.

Avoiding Overfitting

- A variety of techniques to avoid overfitting:
 - Cross-validation
 - Regularization
 - Feature selection
 - Dimensionality reduction

Cross-validation

- In conventional validation
 - Data partitioned into training, validation and test data set
 - Validation set is used to experiment and tune parameters of learnt model or to compare performances of prediction algorithms (created using the training set) and choose the algorithm with best performance.
- Validation can be enough if, even after partition enough training samples are present and need rough performance estimate
 - Otherwise cross-validation preferable

Cross-validation

- Cross validation is a model evaluation method that checks how well a model generalizes to new data
- In one round of cross-validation
 - Data divided into two subsets
 - One for training and one for testing
 - Testing performance is recorded
- Multiple rounds of cross-validation are performed under different partitions
 - Testing results from all rounds are averaged to generate a more accurate estimate of model prediction performance
- Cross-validation helps to identify if model behaves poorly due to problems like over-fitting

Non-exhaustive Cross-validation

- 2 popular cross-validation schemes
 - Non-exhaustive and Exhaustive
- **k-fold** cross validation quite popular
 - Original data split into k equal-sized folds
 - In each trial, one fold becomes testing set, rest become training set
 - Repeat the process k times



Non-exhaustive Cross-validation

- Holdout method
 - Split data into training and testing set numerous times
 - Issue: Some samples may never end up in testing set while others maybe selected multiple times
 - Testing results from all rounds are averaged to generate model prediction performance

Exhaustive Cross-validation

- Leave out one cross validation (LOOCV) [when k in k -fold set to # of data points]
 - Let each datum be in the testing set once
 - For dataset of size n , LOOCV needs n rounds of cross-validation [can be slow when n is large]
 - Benefit of so many fit and evaluated models is a more robust estimate of model performance as each row of data can represent the entirety of the test dataset.